Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	44.09786	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	86.00882	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1709.50804	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.109282	0.109282	1462.0288	0.559805	13.42

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 44.09786.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 86.00882.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1709.50804.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.204197.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.109282.
- Held-out runtime ms: 1462.0288.
- Robustness dispersion across held-out structure families: 0.009168.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.161842, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.109343, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.107204, errors=[]

- X-n247-k50: feasible=True, penalized gap=0.087219, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.080801, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite

Metric	Baseline Nearest Neighbor	Baseline Clarke-Wright	Baseline Regret Insertion LS	Evolved Solver (Phase-9)
Feasibility Rate	100%	100%	100%	100%
Heldout Feasible Gap	0.273	**0.074 (Best)**	0.466	0.109
Heldout Penalized Gap	0.273	0.074	0.466	0.109
Heldout Runtime (ms)	44.1	86.0	1709.5	1462.0
Robustness Dispersion	0.0028	0.0078	0.0869	0.0092

- **Feasibility:** All methods including the evolved solver maintain perfect feasibility (1.0).
- **Objective Gap:** The Clarke-Wright baseline achieves the lowest heldout gaps, outperforming the evolved solver in solution quality by a noticeable margin (0.074 vs. 0.109).
- **Runtime:** The evolved solver is significantly slower than Clarke-Wright and nearest neighbor, though faster than regret insertion local search.
- **Robustness:** The evolved solver shows low robustness dispersion close to Clarke-Wright, indicating consistent performance.
- **Novelty & Complexity:** Evolved solver exhibits high code novelty and moderate complexity, reflecting innovation.

Conclusion

The evolved solver achieves perfect feasibility and reasonable robustness but does **not clearly outperform** the best fixed baseline (Clarke-Wright) in objective gap or runtime on held-out instances across multiple structure families.